

Daily Sequences Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Monotonicity & Boundedness · Fixed Points · Non-Modular Periodicity · Floor/Ceiling Sequences.

Section A Rapid Recognition

1. Is the sequence $a_n = 5 - n$ increasing, decreasing, or neither?

Answer: Decreasing

Method: $a_{n+1} - a_n = (5 - (n + 1)) - (5 - n) = -1 < 0$ for every n .

Investigate further: Every AP with negative common difference is decreasing — this is just Day 1's definition, restated in “increasing/decreasing” language.

2. Is the sequence $a_n = n^2$ bounded above? Bounded below?

Answer: Not bounded above; bounded below (by 1).

Method: $n^2 \rightarrow \infty$ as $n \rightarrow \infty$, so no upper bound exists. But $n^2 \geq 1$ for every positive integer n , so 1 is a lower bound.

Investigate further: A sequence can be bounded on one side only — always check both directions separately.

3. Find the fixed point of $f(x) = 2x - 3$ (i.e. solve $f(x) = x$).

Answer: $x = 3$

Method: $x = 2x - 3 \Rightarrow -x = -3 \Rightarrow x = 3$.

Investigate further: Check: $f(3) = 2(3) - 3 = 3$ ✓ — once a sequence generated by f lands exactly on 3, every later term stays at 3 forever.

4. A sequence satisfies $a_{n+1} = \frac{1}{a_n}$. If $a_1 = 2$, find a_2, a_3, a_4 .

Answer: $a_2 = \frac{1}{2}, a_3 = 2, a_4 = \frac{1}{2}$

Method: $a_2 = \frac{1}{a_1} = \frac{1}{2}$. $a_3 = \frac{1}{a_2} = 2$. $a_4 = \frac{1}{a_3} = \frac{1}{2}$.

Investigate further: This sequence has period 2: odd-indexed terms are 2, even-indexed terms are $\frac{1}{2}$, forever — an early taste of today's periodicity theme.

5. Compute $\lfloor 3.7 \rfloor$ and $\lceil 3.2 \rceil$.

Answer: $\lfloor 3.7 \rfloor = 3$, $\lceil 3.2 \rceil = 4$

Method: Floor rounds down to the nearest integer; ceiling rounds up.

Investigate further: Both operations leave an already-integer value unchanged — $\lfloor 5 \rfloor = \lceil 5 \rceil = 5$.

6. Recall from Day 4 that a recurrence with both characteristic roots inside the unit circle converges to 0. Does $a_n = \left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n$ converge as $n \rightarrow \infty$? To what value?

Answer: Yes, converges to 0.

Method: Both $\left(\frac{1}{2}\right)^n \rightarrow 0$ and $\left(\frac{1}{3}\right)^n \rightarrow 0$ as $n \rightarrow \infty$, since both bases have magnitude less than 1. Their sum also $\rightarrow 0$.

Investigate further: This is literally Day 4's D5 result, applied to specific numbers: $a_n = Ar_1^n + Br_2^n$ with $A = B = 1$, $r_1 = \frac{1}{2}$, $r_2 = \frac{1}{3}$, both inside the unit circle.

7. Is the sequence $a_n = (-1)^n$ bounded? Does it converge?

Answer: Bounded (yes); does not converge.

Method: Every term is either 1 or -1 , so $-1 \leq a_n \leq 1$ always (bounded). But the sequence never settles near a single value — it alternates forever.

Investigate further: This is the standard counterexample to “boundedness implies convergence” — see A8 and Common Trap 1.

8. True or false: every bounded sequence converges.

Answer: False

Method: $a_n = (-1)^n$ (A7) is a direct counterexample: bounded, but not convergent.

Investigate further: The correct combined statement is: bounded *and monotonic* together imply convergence — boundedness alone is not enough.

9. Is the sequence $a_n = \frac{n}{n+1}$ increasing or decreasing? (Check a_1, a_2, a_3 .)

Answer: Increasing

Method: $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$ — each term is bigger than the last.

Investigate further: B1 proves this rigorously for all n , not just the first three terms.

10. Find the smallest n for which $\lfloor \sqrt{n} \rfloor = 4$.

Answer: 16

Method: $\lfloor \sqrt{n} \rfloor = 4$ needs $4 \leq \sqrt{n} < 5$, i.e. $16 \leq n < 25$. The smallest such n is 16.

Investigate further: This is the boundary case of C5's counting question — always convert a floor equation into a double inequality on n before searching.

Section B Manipulation Drills

1. Prove that $a_n = \frac{n}{n+1}$ is increasing for all $n \geq 1$ by showing $a_{n+1} - a_n > 0$.

Answer: Proof: $a_{n+1} - a_n = \frac{1}{(n+1)(n+2)} > 0$.

Method: $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)} = \frac{n^2 + 2n + 1 - n^2 - 2n}{(n+2)(n+1)} = \frac{1}{(n+1)(n+2)}$, which is positive for every $n \geq 1$. $\square$

Investigate further: This “common denominator, then simplify the numerator” technique for proving $a_{n+1} - a_n > 0$ works on almost any rational sequence — practice it on B2’s boundedness question next.

2. Is $a_n = \frac{n}{n+1}$ bounded above? If so, by what value (the supremum)?

- A) Yes, supremum 1.
 B) Yes, supremum 2.
 C) No, unbounded.
 D) Yes, supremum $\frac{1}{2}$.

Answer: A) Yes, supremum 1.

Method: $a_n = \frac{n}{n+1} = 1 - \frac{1}{n+1} < 1$ for every n , and $a_n \rightarrow 1$ as $n \rightarrow \infty$ — so 1 is the smallest possible upper bound (the supremum), even though no term ever equals it exactly.

Investigate further: Rewriting $\frac{n}{n+1}$ as $1 - \frac{1}{n+1}$ makes both the increasing behaviour (B1) and the bound (here) obvious at a glance — always look for this kind of rewrite on a rational sequence.

3. A sequence satisfies $a_{n+1} = \frac{1}{a_n}$ with $a_1 = 3$. Find a_{100} .

- A) 3
 B) $\frac{1}{3}$
 C) 100
 D) $\frac{1}{100}$

Answer: B) $\frac{1}{3}$

Method: The sequence has period 2: $a_1 = 3, a_2 = \frac{1}{3}, a_3 = 3, a_4 = \frac{1}{3}, \dots$ — odd indices give 3, even indices give $\frac{1}{3}$. Since 100 is even, $a_{100} = \frac{1}{3}$.

Investigate further: Always reduce a large index modulo the period before doing anything else — $100 \bmod 2 = 0$, matching the “even” case.

4. Find the fixed point(s) of $f(x) = \frac{1}{x}$ (i.e. solve $f(x) = x$).

Answer: $x = \pm 1$

Method: $\frac{1}{x} = x \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Investigate further: A function can have more than one fixed point — $f(x) = \frac{1}{x}$ has two, unlike A3's $f(x) = 2x - 3$, which has exactly one (any non-identity linear function has at most one).

5. A sequence satisfies $a_{n+1} = f(a_n)$ where f has a fixed point at $x = 5$. If $a_1 = 5$, what is a_n for every n ?

- A) $a_n = 5$ for all n .
- B) $a_n = 5n$.
- C) $a_n \rightarrow 5$ but never equals it.
- D) Cannot be determined.

Answer: A) $a_n = 5$ for all n .

Method: $a_1 = 5$ is exactly the fixed point, so $a_2 = f(5) = 5$, and by the same reasoning every later term stays at 5 forever.

Investigate further: This is the simplest possible case of C8's general theorem: if a sequence *starts* at a fixed point, it never leaves.

6. Which of the following sequences is monotonic (always increasing or always decreasing)?

- A) $a_n = (-1)^n$
- B) $a_n = \sin(n)$
- C) $a_n = \frac{1}{n}$
- D) $a_n = n \bmod 3$

Answer: C) $\frac{1}{n}$

Method: $\frac{1}{n}$ strictly decreases as n increases. A) and B) oscillate without settling into one direction; D) cycles through 1, 2, 0, 1, 2, 0, ... forever.

Investigate further: A sequence is monotonic only if *every* consecutive pair moves the same direction — one exception anywhere disqualifies it, which is why periodic sequences (like D) are essentially never monotonic unless constant.

7. Evaluate $\lfloor -2.3 \rfloor$.

- A) -2
- B) -3
- C) 2
- D) 3

Answer: B) -3

Method: Floor always rounds toward $-\infty$, not toward 0: the greatest integer that is still ≤ -2.3 is -3 , not -2 .

Investigate further: This is the single most common floor-function slip — for negative numbers, “round down” means *more* negative, not closer to zero.

8. A sequence is defined by $a_n = \left\lfloor \frac{n}{2} \right\rfloor$. Find a_1, a_2, a_3, a_4, a_5 .

- A) 0, 1, 1, 2, 2
 B) 1, 1, 2, 2, 3
 C) 0, 0, 1, 1, 2
 D) 0, 1, 2, 2, 3

Answer: A) 0, 1, 1, 2, 2

Method: $a_1 = \lfloor 0.5 \rfloor = 0$, $a_2 = \lfloor 1 \rfloor = 1$, $a_3 = \lfloor 1.5 \rfloor = 1$, $a_4 = \lfloor 2 \rfloor = 2$, $a_5 = \lfloor 2.5 \rfloor = 2$.

Investigate further: This sequence repeats each integer value exactly twice (except possibly the first) — a simple example of how floor division “compresses” a sequence.

9. Determine whether $a_n = \frac{(-1)^n}{n}$ converges, and if so, to what limit.

Answer: Converges to 0.

Method: $\left| \frac{(-1)^n}{n} \right| = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, and since $-\frac{1}{n} \leq a_n \leq \frac{1}{n}$, the sequence is squeezed to 0 regardless of the alternating sign.

Investigate further: This is a genuinely different situation from A7’s $(-1)^n$: here the *magnitude* shrinks to 0 as well as the sign alternating, so the oscillation itself dies out and convergence does happen.

10. For an AP with common difference d , under what condition on d is the sequence strictly increasing?

- A) $d > 0$
 B) $d < 0$
 C) $d \neq 0$
 D) $d \geq 0$

Answer: A) $d > 0$

Method: $a_{n+1} - a_n = d$ for every n in an AP — strictly increasing means this difference must be positive, i.e. $d > 0$.

Investigate further: Compare to Day 1: this is the same fact as Day 1’s toolkit, now phrased in this sheet’s “increasing/decreasing” language rather than “common difference.”

Section C Substitution & Structure

1. A sequence satisfies $a_{n+1} = \frac{a_n - 1}{a_n + 1}$, with $a_1 = 2$. Find a_5 .

- A) 2
 B) $\frac{1}{3}$
 C) $-\frac{1}{2}$

D) -3 **Answer:** A) 2

Method: $a_1 = 2$. $a_2 = \frac{2-1}{2+1} = \frac{1}{3}$. $a_3 = \frac{1/3-1}{1/3+1} = \frac{-2/3}{4/3} = -\frac{1}{2}$. $a_4 = \frac{-1/2-1}{-1/2+1} = \frac{-3/2}{1/2} = -3$.
 $a_5 = \frac{-3-1}{-3+1} = \frac{-4}{-2} = 2$.

Investigate further: This Möbius map has exact period 4: $a_5 = a_1$, so the cycle $2, \frac{1}{3}, -\frac{1}{2}, -3$ repeats forever — a classical fact about the transformation $x \mapsto \frac{x-1}{x+1}$.

2. Continuing C1's sequence, what is a_{101} ?

A) 2

B) $\frac{1}{3}$ C) $-\frac{1}{2}$ D) -3 **Answer:** A) 2

Method: The sequence has period 4 (from C1). $101 = 4(25) + 1$, so $101 \equiv 1 \pmod{4}$, meaning $a_{101} = a_1 = 2$.

Investigate further: Exactly the same index-reduction technique as B3, just with period 4 instead of 2 — always find the period first, then reduce the target index modulo it.

3. A sequence satisfies $a_1 = 1$ and $a_{n+1} = \sqrt{a_n + 2}$. Which is true about (a_n) ?

A) Increasing and bounded above by 2; converges to 2.

B) Decreasing and bounded below by 1.

C) Not monotonic; oscillates.

D) Unbounded, diverges to infinity.

Answer: A) Increasing and bounded above by 2; converges to 2.

Method: Computing terms: $a_1 = 1, a_2 = \sqrt{3} \approx 1.73, a_3 \approx 1.93, a_4 \approx 1.98, \dots$ — visibly increasing and approaching 2. (D1 proves this rigorously.)

Investigate further: The fixed point of $f(x) = \sqrt{x+2}$ solves $x = \sqrt{x+2} \Rightarrow x^2 = x+2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x = 2$ (rejecting $x = -1$ since $f(x) \geq 0$ always) — exactly matching the numerically observed limit.

4. The first seven terms of a sequence of positive integers are: $w_1 = 10, w_2 = 15, w_3 = 18, w_4 = 22, w_5 = 31, w_6 = 36, w_7 = 41$. Consider the statement (*): "If n is prime, then w_n is a multiple of 3 or a multiple of 5." What is the smallest value of n that provides a counterexample to (*)? (after TMUA 2021 Paper 2 Q10)

A) 2

B) 3

C) 5

D) 7

Answer: C) 5

Method: Check the primes in order: $n = 2$: $w_2 = 15 = 3 \times 5$, a multiple of both — not a counterexample. $n = 3$: $w_3 = 18 = 3 \times 6$, a multiple of 3 — not a counterexample. $n = 5$: $w_5 = 31$, which is neither a multiple of 3 ($31 = 3 \times 10 + 1$) nor of 5 — this IS a counterexample. Since 5 is the smallest prime that fails, it's the answer.

Investigate further: The real TMUA question (2021 Paper 2 Q10) uses the sequence 15, 21, 30, 37, 44, 51, 59 and the claim “multiple of 3 or 5,” also landing on $n = 5$ as the smallest counterexample — always check primes in increasing order and stop at the first failure, rather than checking every term first.

5. A sequence is defined by $a_n = \lfloor \sqrt{n} \rfloor$. For how many values of n in the range $1 \leq n \leq 100$ does $a_n = 7$?

- A) 13
B) 14
C) 15
D) 16

Answer: C) 15

Method: $a_n = 7$ needs $7 \leq \sqrt{n} < 8$, i.e. $49 \leq n < 64$. The integers from 49 to 63 inclusive number $63 - 49 + 1 = 15$.

Investigate further: This is A10's boundary-finding technique extended to counting a whole range — always convert to a double inequality on n first, then count the integers in that range.

6. A sequence is defined by $a_n = n - \lfloor n/3 \rfloor \cdot 3$ (the remainder when n is divided by 3, written using floor notation). Is (a_n) periodic? If so, what is its period?

- A) Periodic, period 3.
B) Periodic, period 2.
C) Not periodic.
D) Periodic, period 1 (constant).

Answer: A) Periodic, period 3.

Method: $a_n = n - 3\lfloor n/3 \rfloor$ is exactly the remainder of n divided by 3: for $n = 1, 2, 3, 4, 5, 6, \dots$ this gives 1, 2, 0, 1, 2, 0, $\dots$ — repeating every 3 terms.

Investigate further: Floor notation and the mod operator describe exactly the same thing here — $n \bmod 3$ and $n - 3\lfloor n/3 \rfloor$ are identical formulas written two different ways.

7. A sequence satisfies $a_{n+1} = a_n^2 - a_n + 1$ for $n \geq 1$, with $0 < a_1 < 1$. Is (a_n) increasing, decreasing, or could it be either depending on a_1 ?

- A) Always increasing (or constant), for any starting value.
B) Always decreasing.
C) Depends on a_1 .

D) Cannot be determined without more information.

Answer: A) Always increasing (or constant), for any starting value.

Method: $a_{n+1} - a_n = (a_n^2 - a_n + 1) - a_n = a_n^2 - 2a_n + 1 = (a_n - 1)^2 \geq 0$ always, regardless of a_n 's value — so $a_{n+1} \geq a_n$ for every n , for *any* real starting value, not just $0 < a_1 < 1$.

Investigate further: The restriction “ $0 < a_1 < 1$ ” in the question is a red herring — the algebra shows the result holds unconditionally. Recognising when a stated constraint isn't actually load-bearing is itself a skill (see Common Trap 5).

8. For a sequence defined by $a_{n+1} = f(a_n)$ with f continuous, if (a_n) converges to a limit L , what must be true about L ?

A) $f(L) = L$ — L must be a fixed point of f .

B) $L = a_1$ always.

C) $f(L) = 0$.

D) No general statement can be made.

Answer: A) $f(L) = L$ — L must be a fixed point of f .

Method: Taking the limit of both sides of $a_{n+1} = f(a_n)$ as $n \rightarrow \infty$: the left side $\rightarrow L$, and since f is continuous, the right side $\rightarrow f(L)$. So $L = f(L)$.

Investigate further: This is the theorem underlying every fixed-point computation in this sheet (A3, B4, B5, C3, D1, D4) — it's why solving $f(x) = x$ is always the right first move when a recursively-defined sequence is claimed (or suspected) to converge.

Section D Challenge

1. A sequence satisfies $a_1 = 1$ and $a_{n+1} = \sqrt{a_n + 2}$ (as in C3). Prove rigorously, by induction, that (a_n) is bounded above by 2 for all n ; then prove (a_n) is strictly increasing; and hence explain why the sequence must converge, identifying its limit.

Answer: Bounded above by 2, strictly increasing, converges to 2.

Method: *Bounded above by 2 (induction):* Base case $a_1 = 1 < 2$. Inductive step: assume $a_k < 2$; then $a_{k+1} = \sqrt{a_k + 2} < \sqrt{2 + 2} = 2$. By induction, $a_n < 2$ for all n .

Strictly increasing: Since $0 < a_n < 2$, we have $(a_n - 2) < 0$ and $(a_n + 1) > 0$, so $(a_n - 2)(a_n + 1) < 0$, i.e. $a_n^2 - a_n - 2 < 0$, i.e. $a_n + 2 > a_n^2$. Since $a_n \geq 0$, taking square roots preserves the inequality: $\sqrt{a_n + 2} > a_n$, i.e. $a_{n+1} > a_n$.

Convergence: An increasing sequence bounded above must converge (Monotone Convergence Theorem). Taking the limit L of $a_{n+1} = \sqrt{a_n + 2}$: $L = \sqrt{L + 2} \Rightarrow L^2 = L + 2 \Rightarrow (L - 2)(L + 1) = 0 \Rightarrow L = 2$ or $L = -1$. Since every $a_n > 0$, $L \geq 0$, so $L = 2$. $\square$

Investigate further: This is the complete, rigorous version of what C3 only observed numerically — notice that all three pieces (boundedness, monotonicity, and the limit equation) reuse ideas from earlier in the sheet: induction, the $(a_n - 2)(a_n + 1)$ factoring trick from C7's style of algebra, and C8's fixed-point theorem.

2. A Möbius map $f(x) = \frac{1}{1-x}$ generates a sequence via $a_{n+1} = f(a_n)$. Starting from $a_1 = 2$, find the period of the resulting sequence (the smallest $k > 0$ such that $a_{n+k} = a_n$ for all n).

- A) 2
- B) 3
- C) 4
- D) 6

Answer: B) 3

Method: $a_1 = 2$. $a_2 = \frac{1}{1-2} = -1$. $a_3 = \frac{1}{1-(-1)} = \frac{1}{2}$. $a_4 = \frac{1}{1-\frac{1}{2}} = 2 = a_1$.

Investigate further: This is a different Möbius map from C1/C2's — $f(x) = \frac{1}{1-x}$ has order 3 under composition (a classical fact), while C1's $f(x) = \frac{x-1}{x+1}$ has order 4. Both are genuine, exact cycles — always iterate directly rather than assuming a map is aperiodic.

3. A sequence is defined by $a_1 = 100$ and $a_{n+1} = \lfloor a_n/2 \rfloor$. Find the smallest n for which $a_n = 0$.

- A) 6
- B) 7
- C) 8
- D) 9

Answer: C) 8

Method: $a_1 = 100, a_2 = 50, a_3 = 25, a_4 = 12, a_5 = 6, a_6 = 3, a_7 = 1, a_8 = 0$.

Investigate further: Repeated floor-halving reaches 0 in roughly $\log_2(a_1)$ steps — here $\log_2 100 \approx 6.6$, and the actual count (7 steps from 100 to 0, i.e. $a_8 = 0$) matches that estimate closely, since each floor operation can only lose at most one extra step compared to plain halving.

4. A sequence satisfies $a_1 = 3$ and $a_{n+1} = \frac{a_n + \frac{4}{a_n}}{2}$. Which best describes its behaviour?

- A) Decreasing (after the first term) and bounded below by 2; converges rapidly to 2.
- B) Increasing, converges to 2.
- C) Oscillates between values above and below 2.
- D) Diverges to infinity.

Answer: A) Decreasing (after the first term) and bounded below by 2; converges rapidly to 2.

Method: $a_1 = 3, a_2 = \frac{3 + 4/3}{2} = \frac{13/3}{2} = \frac{13}{6} \approx 2.167, a_3 \approx 2.006, a_4 \approx 2.00002$ — rapidly approaching 2 from above. The fixed point solves $x = \frac{x + 4/x}{2} \Rightarrow 2x = x + \frac{4}{x} \Rightarrow x = \frac{4}{x} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root, since all terms stay positive).

Investigate further: This is Newton's method (the Babylonian method) for computing $\sqrt{4}$ — the same iteration $a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right)$ computes $\sqrt{c}$ for any $c > 0$, converging extremely fast (the number of correct digits roughly doubles every step).

5. A sequence satisfies $a_{n+1} = a_n(1 - a_n)$ with $a_1 = \frac{1}{2}$. Compute a_2, a_3 , and determine whether the sequence is monotonic.

- A) Decreasing for all n , since $a_{n+1} - a_n = -a_n^2 \leq 0$ always.
 B) Increasing for all n .
 C) Not monotonic.
 D) Constant.

Answer: A) Decreasing for all n .

Method: $a_2 = \frac{1}{2} \left(1 - \frac{1}{2}\right) = \frac{1}{4}$. $a_3 = \frac{1}{4} \left(1 - \frac{1}{4}\right) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$. In general, $a_{n+1} - a_n = a_n(1 - a_n) - a_n = a_n(1 - a_n - 1) = -a_n^2 \leq 0$ always, so the sequence never increases.

Investigate further: This is C7's exact technique in reverse-flavour: there, $(a_n - 1)^2 \geq 0$ forced the sequence up; here, $-a_n^2 \leq 0$ forces it down. Both times, rewriting $a_{n+1} - a_n$ as $\pm(\text{something})^2$ settles monotonicity instantly, without needing to compute many terms or invoke boundedness at all.

Top 5 Patterns — Worksheet #5

- Monotonicity via the sign of $a_{n+1} - a_n$.** Often collapses to $\pm(\text{something})^2$, settling the direction instantly without computing terms. *(see A1, A9, B1, C7, D5)*
- Fixed points: solve $f(L) = L$.** If a sequence converges, its limit must be a fixed point of the generating function. *(see A3, B4, B5, C8, D4)*
- Boundedness + monotonicity together imply convergence.** Neither alone is enough — see Common Trap 1. *(see C3, D1)*
- Möbius maps can have small, exact periods under repeated composition.** Check by direct iteration — don't assume a map is aperiodic just because it looks complicated. *(see C1, C2, D2)*
- Floor/ceiling notation is just an explicit “round down/up.”** Expressions like $\lfloor n/k \rfloor$ often secretly encode periodicity (a remainder) or a counting bound. *(see A5, A10, B7, B8, C5, C6, D3)*

Common Traps — Worksheet #5

- Assuming boundedness alone implies convergence.** $(-1)^n$ is bounded but never converges — monotonicity is also needed. *(see A7, A8)*
- Forgetting floor rounds DOWN even for negative numbers.** $\lfloor -2.3 \rfloor = -3$, not -2 . *(see B7)*
- Assuming a sequence must be eventually constant or simply diverging.** Some sequences are genuinely, exactly periodic with period > 1 forever. *(see C1, C2, D2)*
- Trusting a few computed terms as proof of monotonicity.** Numerical evidence is not a proof — always confirm the sign of $a_{n+1} - a_n$ algebraically for all n . *(see D1)*
- Assuming a stated restriction on initial conditions is load-bearing.** Some monotonicity results hold far more generally than a question implies — check whether the algebra actually needs the stated constraint. *(see C7)*

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

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